



CS 693: Compressed Sensing and its applications

Course Code	CS 693
Course Name	Compressed Sensing and its applications
L-T-P-C	3-0-0-3
Prerequisites	Signal Processing
Intended for	Masters/PhD
Distribution	Elective Semester: Odd/Even: Even
Approval	2nd Adhoc Senate (May 2012)

Course Outline

Sparse and Redundant Representations Theoretical and Numerical Foundations , Uniqueness and Uncertainty, Pursuit Algorithms Practice , From Exact to Approximate Solutions, Iterative-Shrinkage Algorithms, The Dantzig-Selector Algorithm, Sparsity-Seeking Methods in Signal Processing, The Quest for a Dictionary, MAP versus MMSE Estimation, Case study: Image Deblurring, Image Denoising and face recognition.

Text and Reference Books

1. Michael Elad, **Sparse and Redundant Representations : From Theory to Applications in Signal and Image Processing**, Springer, 2010.
2. Research articles from IEEE